

# Assignment 3 - OOPs in Python (Solved)

## 1. Student Class

```
class Student:
    def __init__(self, name, age, course):
        self.name = name
        self.age = age
        self.course = course

student1 = Student("Harsh", 20, "Data Science")

print("Name:", student1.name)
print("Age:", student1.age)
print("Course:", student1.course)
```

## 2. Car Class

```
class Car:
    def __init__(self, brand, model, price):
        self.brand = brand
        self.model = model
        self.price = price

car1 = Car("Toyota", "Innova", 2000000)
car2 = Car("Hyundai", "Creta", 1500000)

print(car1.brand, car1.model, car1.price)
print(car2.brand, car2.model, car2.price)
```

## 3. Employee Class

```
class Employee:
    def __init__(self, employee_id, name, salary):
        self.employee_id = employee_id
        self.name = name
        self.salary = salary

emp = Employee(101, "Harsh", 50000)

print(emp.employee_id, emp.name, emp.salary)
```

## 4. Rectangle Area

```
class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        print("Area =", self.length * self.width)

r = Rectangle(10, 5)
r.area()
```

## 5. Circle Area

```
class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        print("Area =", 3.14 * self.radius * self.radius)
```

```
c = Circle(7)
c.area()
```

## 6. BankAccount

```
class BankAccount:
    def __init__(self, account_holder, balance):
        self.account_holder = account_holder
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
        print("Balance:", self.balance)

    def withdraw(self, amount):
        self.balance -= amount
        print("Balance:", self.balance)

acc = BankAccount("Harsh", 10000)
acc.deposit(2000)
acc.withdraw(1500)
```

## 7. Inheritance - Animal and Dog

```
class Animal:
    def sound(self):
        print("Animal Sound")

class Dog(Animal):
    def sound(self):
        print("Bark")

d = Dog()
d.sound()
```

## 8. Person and Student

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

class Student(Person):
    def __init__(self, name, age, roll_number):
        super().__init__(name, age)
        self.roll_number = roll_number

s = Student("Harsh", 20, 101)

print(s.name)
print(s.age)
print(s.roll_number)
```

## 9. Calculator

```
class Calculator:
    def add(self, a, b):
        return a + b

    def subtract(self, a, b):
        return a - b

    def multiply(self, a, b):
        return a * b

    def divide(self, a, b):
        return a / b
```

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

c = Calculator()
print("Add =", c.add(a, b))
print("Subtract =", c.subtract(a, b))
print("Multiply =", c.multiply(a, b))
print("Divide =", c.divide(a, b))
```

## 10. LibraryBook

```
class LibraryBook:
    def __init__(self, book_name, author, price):
        self.book_name = book_name
        self.author = author
        self.price = price

    def display_book_info(self):
        print(self.book_name, self.author, self.price)

b1 = LibraryBook("Python", "Guido", 500)
b2 = LibraryBook("AI Basics", "John", 650)
b3 = LibraryBook("Data Science", "David", 700)

b1.display_book_info()
b2.display_book_info()
b3.display_book_info()
```